

AYMANE AIT DADS

Data Scientist - Neural Net Forecasting | Time Series · Deep Learning · Scalable ML Deployment
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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and executing a rigorous LoRA SFT pipeline on a 30B-parameter model - directly demonstrating the ability to prototype advanced deep learning solutions with minimal support. At Orange Maroc, built ensemble machine learning pipelines on 100K+ network records, reducing manual analysis time by 60% and producing outputs used directly by the network optimization team for large-scale operational decisions. Brings hands-on expertise in neural network design, ablation-driven model evaluation, big data feature engineering, and stakeholder communication of complex modelling outputs - skills mapped precisely to Tesco's forecasting and productionisation requirements. Aims to apply rigorous backtesting methodology, time-aware validation, and scalable deep learning forecasting models to Tesco's retail supply chain and demand planning challenges across physical stores and online channels.

CORE COMPETENCIES

Neural Network Forecasting | Deep Learning Model Design | Rigorous Model Evaluation | Time Series Analysis | Scalable Deployment | Ablation Studies | Stakeholder Communication | Big Data Analytics

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Designed and applied a **Random Forest + K-Means ensemble pipeline** on 100K+ fiber-optic network records (upload speed, latency, jitter) to classify network quality across thousands of nodes at scale.
- Built a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure maintenance decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation, and end-to-end pipeline design covering data ingestion through scored output.
- Delivered stakeholder presentations translating **complex model outputs** into actionable maintenance recommendations for non-expert business audiences, demonstrating clear communication of data science findings.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (Jun 2026) · Top 17 / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that over-expanded reasoning - not incorrect reasoning - was the primary failure mode, informing targeted ablation experiments on LR schedules, synthetic augmentation, and packing strategies.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · vLLM

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning a CLIP ViT-L/14 (428M parameters) backbone with a custom MLP head, dual learning rates, and a 3-model ensemble with 10-view TTA across 250K training images and 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered that 1-epoch training (0.793 AUC) outperformed 4-epoch training (0.767 AUC), confirming that early stopping preserves out-of-distribution generalisation in large pretrained vision models.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

Deep Learning & Forecasting: PyTorch | Neural Network Design | Ablation Studies | Ensemble Methods | FP16/BF16 Mixed Precision | TensorFlow | Scikit-learn

Data Science & Analytics: Python | SQL | Pandas | NumPy | Feature Engineering | Clustering | Power BI

MLOps & Infrastructure: Git | Linux | Docker | Kaggle GPU | CUDA | REST APIs | Hugging Face Transformers